SOLUTIONS: STABILITY AND CONTROL OF NONLINEAR SYSTEMS
MSC IN CONTROL & OPTIMISATION

1. Exercise

a) The vector field $f(x_1,x_2)$ is smooth (i.e. $\mathscr{C}^\infty$) with respect to both variables. Hence it is a locally Lipschitz function and by Caratheodory's theorem solutions exist and are unique (over their maximal time-interval of definition). [2]

b) The nullclines of the vector-field can be computed as shown below:

$$\mathscr{N}_1 = \{x : (6-x_1)(6+x_1)x_1 = 0\} = \{x : x_1 = 6\} \cup \{x : x_1 = 0\} \cup \{x : x_1 = -6\}.$$

For the second nullcline we have:

$$\mathscr{N}_2 = \{x : x_1 = x_2^3 - x_2\}.$$

Its local minima and maxima are found for $\frac{d}{dx_2}x_2^3 - x_2 = 3x_2^2 - 1 = 0$, viz $x_2 = \pm 1/\sqrt{3}$. [2]

c) Notice that whenever $x_1(0) = \pm 6$, or $x_1(0) = 0$, we have that the x_1 equation is at equilibrium. Hence $x_1(t) = x_1(0)$ for all t. This shows that, regardless of $x_2(t)$, $x(t) \in \mathscr{N}_1$. [2]

d) There are 10 regions delimited by the nullclines. The direction of the vector-field in each one of them is shown in Fig. 1.1. [2]

e) From the direction of the arrows we notice that no solutions can cross $\mathscr{N}_1$. The regions labeled as 2,4,5,7 are forward invariant. [2]

f) Nullclines intersect in 5 points. In particular, for $x_1 = 6$, we have $x_2 = 2$. For $x_1 = -6$ we have $x_2 = -2$, and for $x_1 = 0$ we have 3 intersections in $x_2 = -1, 0$ and 1 respectively. To linearize the dynamics we first compute the Jacobian matrix $\frac{\partial f}{\partial x}$ at an arbitary value of x.

$$\frac{\partial f}{\partial x}(x) = \begin{bmatrix} 36 - 3x_1^2 & 0 \\ 1 & 1 - 3x_2^2 \end{bmatrix}.$$

At the equilibria $x_e = \pm[6,2]'$ we see that:

$$\frac{\partial f}{\partial x}(x_e) = \begin{bmatrix} -72 & 0 \\ 1 & -11 \end{bmatrix}.$$

The eigenvalues are in -72 and -11, hence two real eigenvalues of negative real part correspond to the phase portrait of a stable node (hyperbolic). On the other hand, for $x_e = [0,0]'$ we have:

$$\frac{\partial f}{\partial x}(x_e) = \begin{bmatrix} 36 & 0 \\ 1 & 1 \end{bmatrix}.$$

Two positive real eigenvalues correspond to the phase portrait of an unstable node (hyperbolic). Finally, for the equilibria in $x_e = \pm[0,1]'$ we see:

$$\frac{\partial f}{\partial x}(x_e) = \begin{bmatrix} 36 & 0 \\ 1 & -2 \end{bmatrix}.$$

Real eigenvalues of opposite signs correspond to a saddle-point. [4]

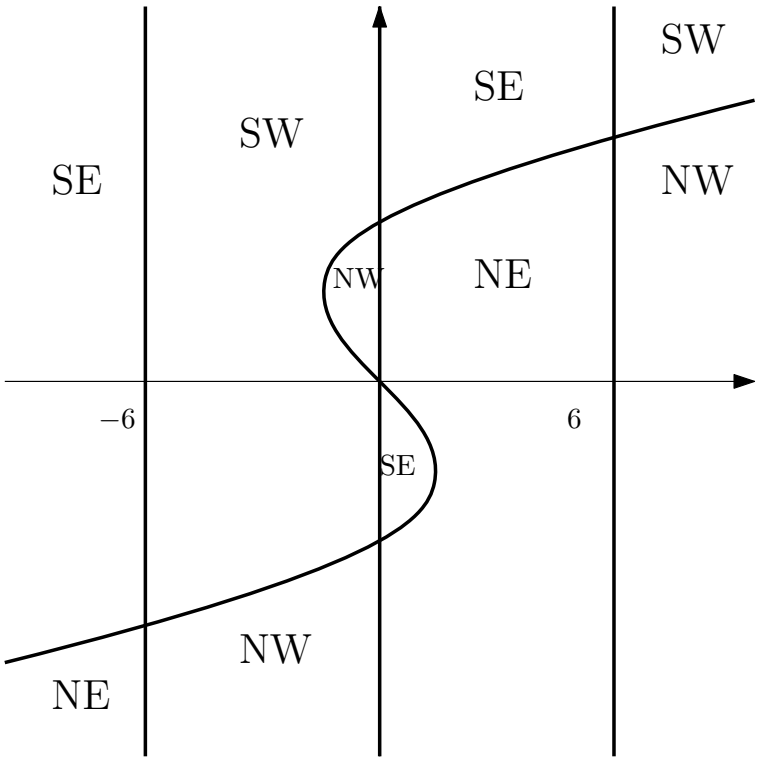


Figure 1.1 Nullclines and vector-field orientations

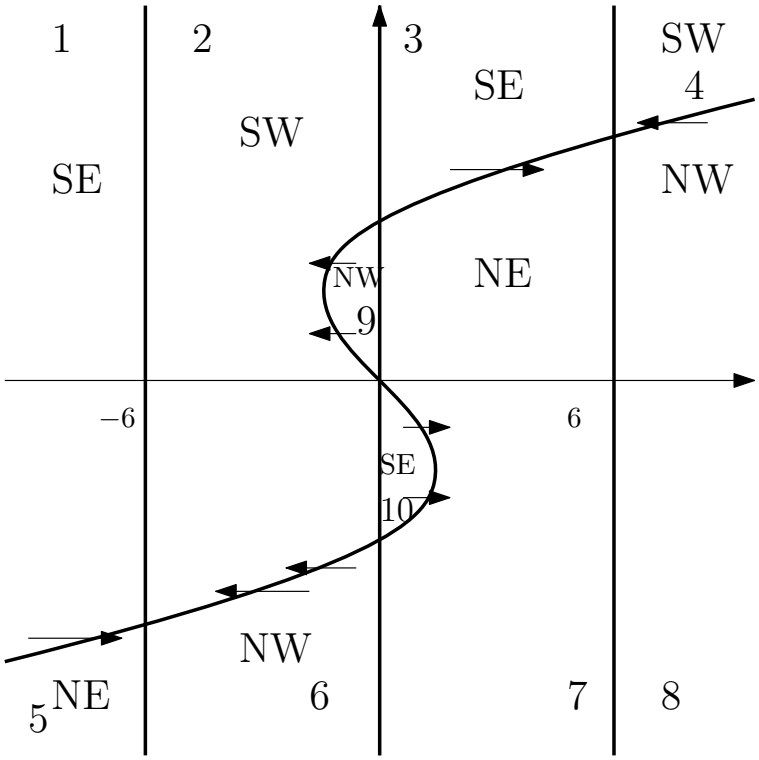


Figure 1.2 Invariant regions

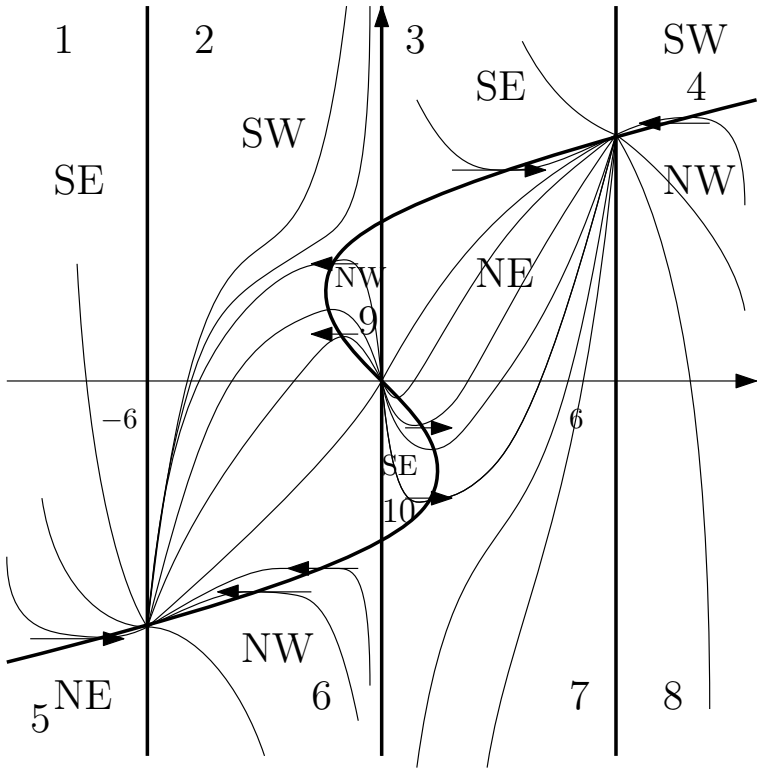


Figure 1.3 Global phase portrait

g) A sketch of the global phase portrait is shown in Fig. 1.3. [3]

h) We can see that every solution converges to an equilibrium. In particular, for $x_1(0) < 0$, $\omega(x_0) = \{[-6,-2]'\}$. For $x_1(0) > 0$, $\omega(x_0) = \{[6,2]'\}$. For $x_1(0) = 0$, $\omega(x_0) = \{[0,1]\}$ if $x_2(0) > 0$ and $\omega(x_0) = \{[0,-1]\}$ if $x_2(0) < 0$. Finally if $x_0 = [0,0]'$, then $\omega(x_0) = \{x_0\}$. [3]

2. Exercise

a) We consider the state variable $x = [z, \dot{z}]'$. Accordingly the following representation is achieved:

$$\dot{x} = \begin{bmatrix} x_2 \\ -[2+\sin(x_1)]x_1 - x_2 + u \end{bmatrix}, \qquad y = x_2.$$

[3]

b) The system is input-affine $\dot{x} = f(x) + g(x)u$, with:

$$f(x) = \begin{bmatrix} x_2 \\ -[2+\sin(x_1)]x_1 - x_2 \end{bmatrix}, \qquad g(x) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

[2]

c) For input-affine systems we know that any storage function for passivity needs to fulfill: $L_g S(x) = y$. Hence, in this case:

$$\frac{\partial S}{\partial x}(x)g(x) = \frac{\partial S}{\partial x_2}(x) = x_2 = y.$$

Integrating with respect to x_2 the previous equality yields:

$$S(x) = \frac{x_2^2}{2} + P(x_1)$$

where $P(x_1)$ is an arbtirary additive integration constant which can be taken to be a function of x_1. [3]

d) Computing the derivative of $S(x)$ as given above, yields:

$$\dot{S} = P'(x_1)x_2 + x_2\dot{x}_2 = x_2[P'(x_1) - [2+\sin(x_1)]x_1 - x_2 + u] = -x_2^2 + yu$$

where the last equality follows if:

$$P'(x_1) = [2+\sin(x_1)]x_1.$$

Hence we can take the storage function as:

$$S(x) = \frac{x_2^2}{2} + \int_0^{x_1} [2+\sin(\xi)]\xi \, d\xi.$$

This yields $\dot{S} \leq yu$ and proves passivity. [3]

e) Stability of the origin can be studied for the closed-loop system with $u = 2y = 2x_2$ by considering the linearized dynamics. In particular the Jacobian at a generic state x equals:

$$J(x) = \begin{bmatrix} 0 & 1 \\ -[2+\sin(x_1)] - \cos(x_1)x_1 & 1 \end{bmatrix}.$$

Linearizing around the origin we have:

$$J(0) = \begin{bmatrix} 0 & 1 \\ -2 & 1 \end{bmatrix}$$

This is a matrix with characteristic polynomial $s^2 - s + 2$, and complex conjugates eigenvalues with positive real part. Hence, the origin is unstable. [3]

f) We prove stability through the Lasalle's criterion. In particular we take:

$$V(x) = \frac{x_2^2}{2} + \int_0^{x_1} [2+\sin(\xi)]\xi \, d\xi.$$

[2]

This is positive definite, since both $x_2^2/2$ and $\int_0^{x_1}[2+\sin(\xi)]\xi d\xi$ are non-negative. Moreover they both vanish if and only if $x = 0$. Radial unboundedness of $V(x)$ is seen by showing boundedness of the set:

$$S_M := \{x : V(x) \leq M\},$$

for arbitrary $M > 0$. In particular, for the integral term we see that $[2+\sin(x_1)] \geq 1$ for all x_1. This implies:

$$\int_0^{x_1} [2+\sin(\xi)]\xi \, d\xi. \geq \int_0^{x_1} \xi d\xi \geq x_1^2/2.$$

Hence, $V(x) \leq M \Rightarrow x_2^2/2 \leq M \Rightarrow |x_2| \leq \sqrt{2M}$. [2]
Similarly, for x_1

$$V(x) \leq M \Rightarrow \int_0^{x_1} [2+\sin(\xi)]\xi d\xi \leq M \Rightarrow x_1^2/2 \leq M \Rightarrow |x_1| \leq \sqrt{2M}.$$

Overall we see that S_M is bounded as $x \in S_M \Rightarrow |x| \leq 2\sqrt{2M}$. Next, compute the derivative of V,

$$\dot{V}(x) = \frac{\partial V}{\partial x}(x) \cdot [f(x) - 2x_2 g(x)] = -3x_2^2.$$

This is negative semi-definite. Moreover:

$$\text{Ker}[\dot{V}] = \{x : x_2 = 0\}.$$

Any invariant limit set contained in the Kernel of $\dot{V}$ is also contained in:

$$\{x : x_2 = 0 \text{ and } \dot{x}_2 = 0\} = \{x : x_2 = 0 \text{ and } -[2+\sin(x_1)]x_1 - 3x_2 = 0\} = \{0\}.$$

This shows that the global Lasalle's stability criterion can be applied and that the origin is GAS. [2]

3. Exercise

a) We consider the following scalar systems:

$$\Sigma_1: \qquad \dot{x}_1 = -3x_1 + \text{atan}(d_1),$$

$$\Sigma_2: \qquad \dot{x}_2 = \text{atan}(-x_2 + 2d_2),$$

along with the interconnections $d_1 = x_2$ and $d_2 = x_1$ which define a feedback loop. [4]

b) We study ISS of Σ_1 by means of the Lyapunov function $V_1(x_1) = x_1^2/2$. This is positive definite and radially unbounded with class $\mathscr{K}_\infty$ lower and upper bounds $\alpha_1(|x_1|) = \alpha_2(|x_1|) = |x_1|^2/2$. Taking derivatives along solutions of Σ_1 yields:

$$\dot{V}_1 = x_1(-3x_1 + \text{atan}(d_1)).$$

For any $\varepsilon > 0$ we have:

$$|x_1| \geq |d_1|/(3-\varepsilon) \Rightarrow \dot{V}_1 \leq -\varepsilon|x_1|^2 - (3-\varepsilon)|x_1|^2 + |x_1||d_1| \leq\leq -\varepsilon|x_1|^2,$$

where we exploited the inequality $\text{atan}(|d_1|) \leq |d_1|$. Notice that $-|x_1|^2$ is negative definite. Hence, choosing $\chi_1(r) := r/(3-\varepsilon)$ we see that V_1 is an ISS-Lyapunov function of type 2 as it fulfills the following implication:

$$|x_1| \geq \chi_1(|d_1|) \Rightarrow \dot{V}_1 \leq -\rho_1(|x_1|)$$

for some positive definite function ρ_1. Hence, Σ_1 is ISS with gain:

$$\gamma_1(r) = \alpha_1^{-1} \circ \alpha_2 \circ \chi_1(r) = \chi_1(r) = r/(3-\varepsilon).$$

[4]

c) We study ISS of Σ_2 by means of the Lyapunov function $V_2(x_2) = x_2^2/2$. This is positive definite and radially unbounded with class $\mathscr{K}_\infty$ the same lower and upper bounds $\alpha_1(|x_2|) = \alpha_2(|x_2|) = |x_2|^2/2$. Taking derivatives along solutions of Σ_2 yields:

$$\dot{V}_2 = x_2\text{atan}(-x_2 + 2d_2).$$

Choosing $|x_2| \geq 2|d_2|/(1-\varepsilon)$, for any sufficiently small $\varepsilon > 0$ yields:

$$|x_2| \geq 2|d_2|(1-\varepsilon) \Rightarrow \dot{V}_2 = x_2\text{atan}(-x_2+2d_2) \leq x_2\text{atan}(-\varepsilon x_2) = -\varepsilon|x_2|\text{atan}(|x_2|).$$

Notice that $-|x_2|\text{atan}(\varepsilon|x_2|)$ is negative definite. Hence, choosing $\chi_2(r) := 2r/(1-\varepsilon)$ we see that V_2 is an ISS-Lyapunov function of type 2 as it fulfills the following implication:

$$|x_2| \geq \chi_2(|d_2|) \Rightarrow \dot{V}_2 \leq -\rho_2(|x_2|)$$

for some positive definite function ρ_2. Hence, Σ_2 is ISS with gain:

$$\gamma_2(r) = \alpha_1^{-1} \circ \alpha_2 \circ \chi_2(r) = \chi_2(r) = 2r/(1-\varepsilon).$$

[4]

d) GAS at the origin of the overall system follows by virtue of the ISS Small Gain Theorem, in particular noticing that:

$$\gamma_1(\gamma_2(r)) = \frac{2}{(1-\varepsilon)(3-\varepsilon)} r < r$$

for all ε sufficiently small and $r > 0$ (pick, for instance, $\varepsilon = 0.1$). This shows that the small-gain condition is satisfied.

[4]

e) Changing $-x_2$ with $+x_2$ in the equation for $\dot{x}_2$ will prevent application of the small gain Theorem. In particular, Σ_2 will no longer be ISS. Notice that its equation, even for $x_1 \equiv 0$ reads: $\dot{x}_2 = \text{atan}(x_2)$ which has an unstable equilibrium in 0, as it follows by the Lyapunov instability criterion applied with Lyapunov function $V(x_2) = x_2^2$, as shown below.

Notice that $V(x_2)$ is positive definite (hence 0 is an accumulation point of $\{x_2 : V(x_2) > 0\}$, and in addition $\dot{V}(x_2) = 2x_2\text{atan}(x_2)$ is also positive definite. This proves instability (in fact anti-stability) of the equilibrium $x_2 = 0$. (A similar conclusion could be reached through linearisation). [4]

4. Exercise

a) To compute the relative degree we start taking derivatives of the output signal.

$$y = x_1, \qquad \dot{y} = x_2 + \text{atan}(x_3)$$

$$\ddot{y} = \dot{x}_2 + \frac{\dot{x}_3}{1+x_3^2} = -2x_2 + x_1^3 + \frac{\cos(x_2)+u}{1+x_3^2}.$$

Notice,

$$\frac{\partial}{\partial u}\ddot{y} = \frac{1}{1+x_3^2} \neq 0 \qquad \forall x$$

Hence 2 is the relative degree and it is globally defined; [3]

b) The state-feedback can be chosen as:

$$u = (1+x_3^2)[v + 2x_2 - x_1^3] - \cos(x_2).$$

This achieves I-O linearization and an equation $\ddot{y} = v$. [3]

c) We choose the following vector $\xi = [y, \dot{y}]^T$, while $z := x_2$. Accordingly:

$$\begin{aligned} \dot{\xi}_1 &= \xi_2 \\ \dot{\xi}_2 &= v \\ \dot{z} &= -2z + y^3 \\ \\ y &= \xi_1 \end{aligned}$$

It is easy to verify that z is only affected by y in its own update equation $\dot{z} = -2z + y^3$. Hence, the latter is the equation of the system in normal form and the internal dynamics are described by the equation $\dot{z} = -2z + y^3$. [6]

d) If we regard y as an exogenous disturbance, by letting $y = d$, the internal dynamics have the equation:

$$\dot{z} = -2z + d^3.$$

This is the equation of an ISS system, as it is easily verified by choosing $V(z) = z^2/2$.

$$\dot{V} = z\dot{z} = -2z^2 + zd^3 \leq -|z|^2 - |z|^2 + |z||d|^3.$$

Accordingly:

$$|z| \geq |d|^3 \Rightarrow \dot{V} \leq -|z|^2,$$

showing that V is an ISS-function of type 2 and the internal dynamics are therefore ISS, as claimed. [3]

e) A PD controller is adequate to stabilize a double integrator. Hence, we choose: $v == -k_p y - k_d \dot{y}$ for some k_p and k_d positive. The resulting closed-loop system is the cascade of the asymptotically stable linear system $\ddot{y} = -k_p y - k_d \dot{y}$ with the internal dynamics. Hence, it is GAS at the equilibrium in the origin. [2]

f) The analysis would change because the internal dynamics would read: $\dot{z} = 2z + y^3$. This is an unstable internal dynamics, and exponentially unstable zero-dynamics, $\dot{z} = 2z$. In particular, stabilization of the y-subsystem does not imply stabilization of the overall system. [3]